

Meanings and Measurements: Multi-Agent Probabilistic Grounding for Vision-Language Navigation

Swagat Padhan^{*,1}, Lakshya Jain^{*,1}, Bhavya Minesh Shah¹, Omkar Patil¹, Thao Nguyen², Nakul Gopalan¹

¹Arizona State University ²Haverford College

{spadhan, ljain6, bshah43, opatil3, ngopalan}@asu.edu, tnguyen3@haverford.edu

Abstract—Robots collaborating with humans must convert natural language goals into actionable, physically grounded decisions. Executing “go two meters to the right of the fridge” requires simultaneously resolving a semantic referent, interpreting a spatial predicate in a geometrically consistent frame, and enforcing an exact metric offset in 3D space. We show empirically that existing VLM-based systems struggle with this joint requirement: scene-graph EQA approaches commit to a single target without metric enforcement, while single-image spatial specialists produce no actionable allocentric goals. To address this, we propose MAPG (Multi-Agent Probabilistic Grounding), a framework that treats metric-semantic goal grounding as compositional inference over 3D space. MAPG decomposes a query into structured clauses, resolves referents via multi-view evidence accumulation against an online 3D scene graph, and instantiates three analytic kernels per clause (semantic, metric, and directional) whose parameters are predicted by specialist VLMs. Kernels compose via a Product of Experts (log-space addition) into a continuous goal distribution $P(x)$ over free navigable space, directly consumable by a motion planner. We introduce MAPG-Bench (98 queries, 41 HM3D scenes), the first benchmark targeting metric-semantic goal grounding, and evaluate on existing embodied QA benchmarks to show competitive performance even when metric constraints are absent. In a completed end-to-end MAPG-Bench run, MAPG succeeds on 79 of 98 queries (80.6%) and finds the anchor on 80 (81.6%). Separate component evaluation reduces object-to-world grounding error from 5.82m to 0.07m (98.8%). A real-world robot demonstration illustrates the approach outside simulation when a structured scene representation is available.

I. INTRODUCTION

Robots that collaborate with humans must be able to convert natural language goals into actionable, grounded decisions. People often mix metric and semantic specifications - from “being five minutes late” to “taking a left after about a mile on the highway.” We refer to these as metric-semantic queries. The semantics relate to the attributes of an object or concept and spatial relationships such as left, right, front, behind; whereas the metric part of the query relates to measurable quantities such as distances, time and scale. Grounding these metric-semantic queries in a map requires jointly reasoning over referent semantics, spatial predicates, and metric scale, where the intended goal is a continuous region anchored to a grounded object pose.

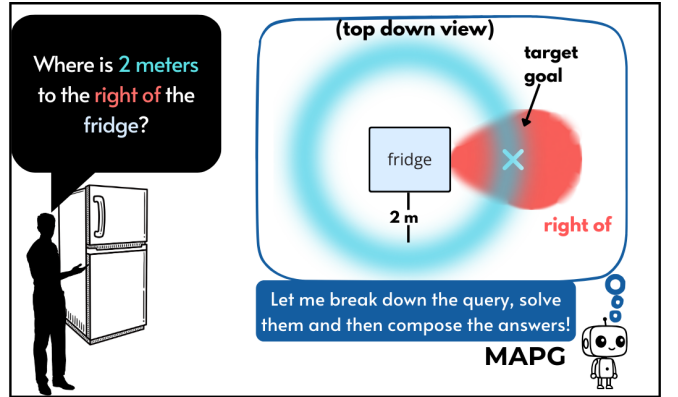


Fig. 1: When a human specifies an open world metric-semantic query, referring to a location in the world, MAPG decomposes the query into subparts as learned probability distributions and composes them for grounding.

Instruction following is a central subfield of grounded language learning in robotics [1]. Most existing systems ground language to actions or goals, which can be objects in a scene [2]. Several grounded language systems emphasized structured decomposition and probabilistic inference to connect linguistic constituents to world entities [3] while modern navigation and embodied question answering (QA) systems increasingly rely on large vision-language models (VLMs) and large language models (LLMs) to reason over observations and internal scene representations such as 3D semantic scene graphs [4], [5]. These approaches treat goal grounding as a single-step decision: given the current observation (and sometimes an internal map), the model is prompted to output either a discrete action or a single target hypothesis [6], [7]. However, such designs can be fragile for metric-semantic instructions whose correctness depends on precise geometry and a consistent frame of reference while navigating to discover the goal. Furthermore, language grounding is bidirectional: the agent must convert egocentric observations into allocentric positions on the map, then convert an allocentric goal back into egocentric coordinates for execution, which compounds errors from each step.

We propose MAPG (Multi-Agent Probabilistic Grounding), an agentic framework that queries different VLM Agents to produce probabilistic distributions and composes them into the goal. Given an instruction, MAPG decomposes

*Equal contribution. This work was partly funded by TSMC and Arizona State University.

it into structured clauses, resolves referents against an online 3D scene graph and the current egocentric view, and instantiates analytic kernels for semantic, metric, and spatial constraints. These kernels are then composed to produce a final goal density that better represents the spatial intention, which a planner can then query to generate executable waypoints. Fig. 1 shows an example of this approach.

Our main contributions are as follows:

- **A multi-agent probabilistic 3D spatial reasoning framework for goal grounding** that couples online 3D scene graphs with analytically defined continuous spatial kernels to produce planner-ready goal distributions for metric-semantic instructions.
- **A first-of-its-kind HM3D [8] goal grounding benchmark for metric-semantic queries**, including an open-source dataset and evaluation protocol spanning 41 unique indoor scenes and 98 annotated metric-semantic queries designed for object-to-world goal grounding in realistic indoor layouts.
- **Empirical findings and ablations:** In a completed 98-query end-to-end evaluation, MAPG succeeds on 79 queries (80.6%) and finds the anchor on 80 (81.6%). We additionally report predicate-level and target-distance breakdowns and provide a failure taxonomy for reproducible comparison.

Our webpage includes videos, benchmark examples, and extensive results: sites.google.com/view/mapg-web/home.

II. BACKGROUND AND RELATED WORK

Our work lies at the intersection of embodied vision-language navigation, structured 3D spatial memory, spatial reasoning with multimodal foundation models, and probabilistic grounding. We review the most relevant directions and highlight the gaps our method addresses.

Embodied Question Answering and Vision-Language Navigation: Embodied Question Answering (EQA) and Vision-and-Language Navigation (VLN) both require an agent to ground language into actions over long horizons under partial observability, using only egocentric observations [9]. Many embodied systems still commit to a single action or discrete target at each step, so early grounding errors compound and can derail the entire trajectory [10], [11], [2]. Some prior VLN-based works explicitly identify this issue and introduce backtracking and progress-based recovery to mitigate trajectory drift under greedy decisions [12]. Conceptually, our approach addresses the same failure mode by shifting goal grounding to producing a planner-ready goal representation in a generated map rather than a one-shot action choice.

Structured 3D Spatial Memory and Scene Graphs: Embodied reasoning benefits from having an explicit spatial memory that persists across time and supports queries about objects, places, and their relations. Hydra [13], along with Kimera [14] construct and optimize an online 3D scene graph that unifies metric layers (e.g., distance fields), object nodes, and higher-level place and room structure [13]. However, scene graphs cannot produce planner-ready navigation goals

as they contain only adjacency information while lacking metric information about a physical space. We therefore build on Scene Graphs a spatial backbone over which metric-semantic language can be composed into planner-ready target distributions (see Fig. 2).

Spatial Reasoning Limits of Multimodal Foundation Models:

Although LLMs and VLMs are strong on many language tasks, their spatial competence remains inconsistent [6], [7], [15], [16]. Across evaluations, common failure modes include unstable visual distance estimates, inconsistent camera framing, and lack of mechanisms to ground visual information [7], [16]. Several works attempt to address these issues through spatial fine-tuning or by incorporating depth, but the gap remains for embodied settings where the agent must maintain a consistent allocentric representation over time [6], [15], [16]. Our approach addresses these limitations at the interface level by making the spatial target explicit within a metric semantic map.

Vision-Language-Action Models and Affordance Learning:

Vision-language-action (VLA) models learn end-to-end policies from multimodal inputs to actions, and large-scale training has shown strong transfer, with navigation-focused variants improving robustness through flexible goal conditioning on language, goal images, or goal poses [17], [18]. However, these policies are typically trained to predict actions rather than to explicitly infer a metric goal in a shared 3D frame. Our work is novel in that it creates compositional interfaces between a planning or navigation stack, a semantic perception stack, and a mapping stack to generate actions.

Probabilistic Grounding and Multi-Expert Fusion:

Grounding language in the physical world has long been treated as probabilistic inference, where linguistic structure induces latent grounding variables over referents, relations, and actions [3], [19]. This formulation is well suited to spatial language, where predicates such as “left of” or “near” define distributions over regions of space rather than point locations. A complementary theme is decomposing a complex task into reusable parts to generalize better than monolithic policies when the combinatorics grow [20], [21], [22]. To combine multiple information sources, products of experts (PoE) offer a principled mechanism that multiplies distributions and renormalizes, producing solutions that satisfy multiple constraints simultaneously [23]. Finally, recent embodied systems increasingly combine multiple models or signals, such as mixing perception with language reasoning or fusing goal proposals with feasibility checks [5].

Our work builds on these directions by addressing the missing interface between structured 3D spatial memory and planner execution for metric-semantic queries, providing a map-aligned and controllable mechanism for translating language into navigation targets (see Fig. 2).

III. METHODOLOGY

Why an agentic decomposition? Multi-agent systems offer a natural interface when subproblems require specialist reasoning whose outputs must be combined into a shared

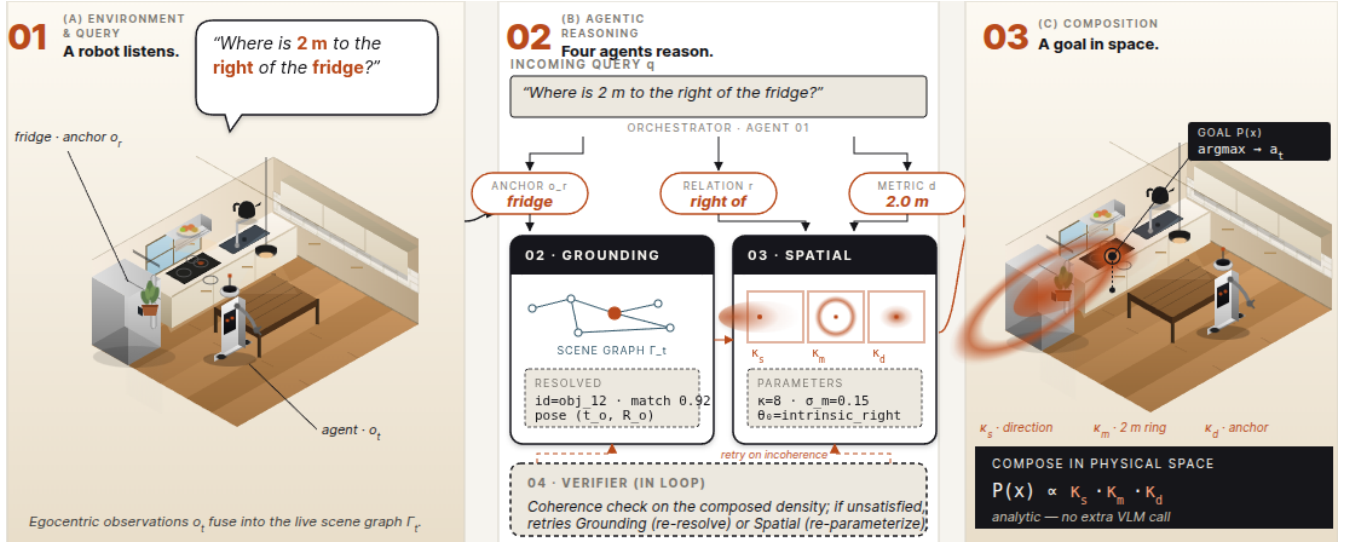


Fig. 2: **MAPG overview.** (a) The robot observes a 3D environment and receives a natural-language spatial query q . (b) *Agentic spatial reasoning layer*: the Orchestrator decomposes q into anchor, relation, and metric tokens $\langle o_r, r, d \rangle$; the Grounding Agent resolves o_r against the 3D scene graph Γ_t using multi-view evidence; the Spatial Agent emits parameters for the directional, metric, and anchor-density kernels. The Verifier closes a retry loop on incoherent compositions. (c) *Composition layer*: MAPG composes the analytic kernels in physical space to produce a continuous target-location PDF $P(x)$, planner-ready as a navigation or object-selection goal.

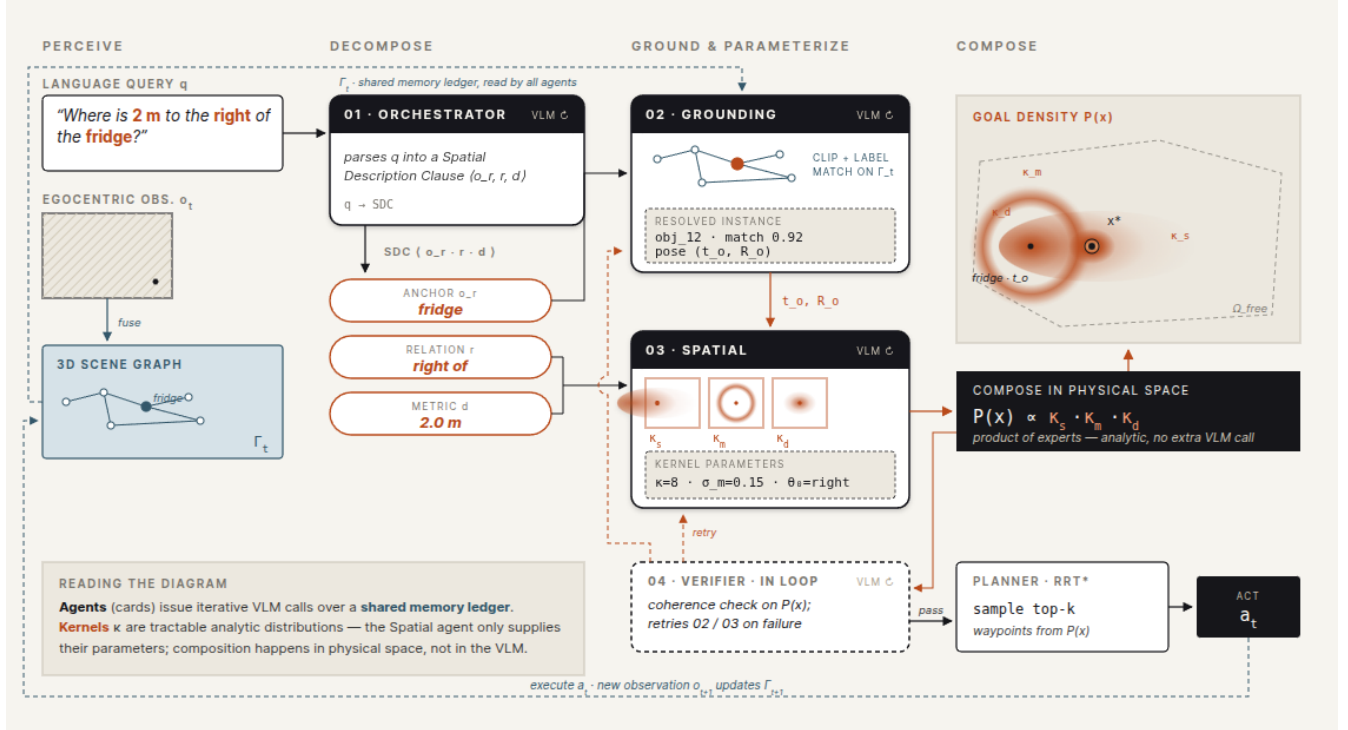


Fig. 3: **MAPG system overview.** Egocentric observations o_t are fused into a 3D scene graph Γ_t , shared by all agents over a memory ledger. The Orchestrator parses the query q into a Spatial Description Clause $\langle o_r, r, d \rangle$; the Grounding Agent resolves the referent o_r against Γ_t ; the Spatial Agent supplies parameters for the directional (κ_s), metric (κ_m), and anchor-density (κ_d) kernels. The kernels are composed analytically in physical space (a product of experts, no extra VLM call), yielding a goal density $P(x)$ over free space. The Verifier checks coherence and triggers corrective retries; on pass, the Planner samples top- k waypoints to act, and the new observation updates Γ_{t+1} , closing the loop.

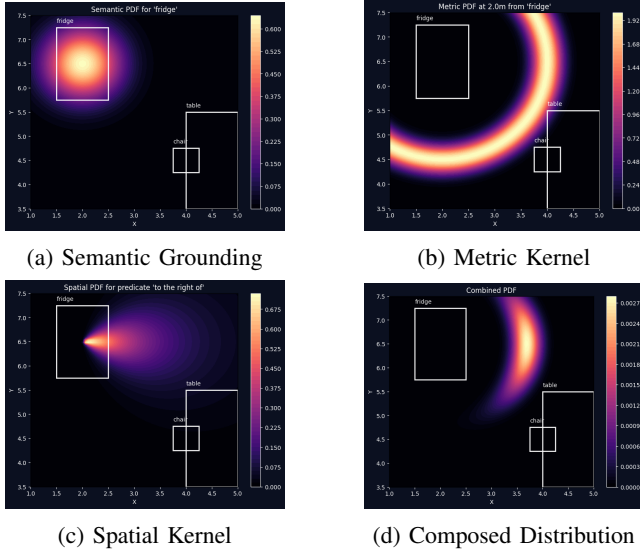


Fig. 4: MAPG grounding for the query “Where is 2m to the right of the fridge?” Semantic grounding (a) identifies the fridge instance; the metric kernel (b) enforces the 2m radial offset; the directional kernel (c) captures “right of” as a vMF distribution in the fridge’s local frame. Their log-space composition (d) yields a planner-ready goal distribution peaking 0.07 m from the ground truth.

representation. Prior work combines agent outputs via language [24] or gradients [21]. Metric-semantic goal grounding presents a unique opportunity: each subproblem (referent resolution, directional reasoning, metric estimation) produces a distribution over *physical space*, so outputs compose directly as a Product of Experts [23], [22], with no learned combiner required. Pre-specifying analytic kernel families mirrors the input-decomposition principle of [25]: each specialist VLM answers only the sub-question it handles best, concentrating capacity where it matters. We use *agent* to mean a module that (a) receives a distinct subproblem, (b) iterates over evidence before returning, and (c) interfaces via structured output. The Orchestrator, Grounding Agent, and Verifier are genuine agents by this definition; the Spatial Kernel Estimator is a single-call parameterized kernel estimator.

Problem Definition. Given a metric-semantic instruction q and a partial 3D scene graph $\Gamma = (V, E)$ with object nodes V , MAPG operates in two modes: *object-to-world* (O-W), which infers a probability density $P(x | q, \Gamma)$ over allocentric waypoints $x \in \Omega_{\text{free}} \subset \mathbb{R}^3$; and *object-to-object* (O-O), which selects among candidate object instances in V . In both modes, high-density regions correspond to locations satisfying the semantic referent, spatial predicate, and metric constraint specified in q . MAPG agents communicate through a shared blackboard storing the scene graph Γ , a referent belief B_t , an event ledger, and global failure history used for closed-loop refinement. MAPG comprises the following components (Fig. 3), drawing on the G³ probabilistic grounding framework [3]:

1) **The Orchestrator** parses a free-form instruction into Spatial Description Clauses (SDCs) [26]: a target entity, anchor objects with optional modifiers and explicit metric distances, and a *composition logic* label. Crucially, the Orchestrator classifies each spatial predicate as either camera-relative (near, between) or intrinsic object-centric (intrinsic.front, intrinsic.back, intrinsic.left, intrinsic.right), enabling downstream kernel computation to select the correct reference frame. For “Where is 2m to the right of the fridge?” it produces: anchor=fridge, composition_logic=intrinsic.right, metric=2.0m. Multi-clause instructions yield a conjunction of SDCs, each resolved independently and composed downstream.

2) **The Grounding Agent** resolves each anchor label to a concrete object instance in Γ . Given a referent like “fridge,” it combines: (i) semantic equivalence matching (“couch” = “sofa”) against scene-graph labels, (ii) visual verification via a VLM query over base64-encoded egocentric images, and (iii) a spatial salience prior (proximity, visibility). Candidates rejected by the Verifier are blacklisted in the shared failure history and excluded from subsequent queries. A belief distribution $B_t(o)$ over candidate instances is maintained on the blackboard; the agent iterates across multiple views until sufficient confidence is reached, making it a genuine evidence-accumulating agent.

3) **Spatial Kernel Estimator** (Spatial Agent): Once anchor o_r with pose $t_o \in \mathbb{R}^3$ is resolved, MAPG instantiates three analytic kernels whose log-likelihoods sum in closed form (Fig. 4):

Semantic kernel: a 3D Gaussian centered at the anchor: $\ell_{\text{sem}}(x) = -\frac{1}{2\sigma_s^2} \|x - t_o\|^2$.

Metric kernel: a radial Gaussian over distance from the anchor: $\ell_{\text{met}}(x) = -\frac{1}{2\sigma_m^2} (\|x - t_o\| - d_0)^2$, where d_0 is parsed from the query by the Orchestrator and σ_m is a system tolerance.

Spatial (directional) kernel: a von Mises-Fisher distribution [3] in the anchor’s local frame: $\ell_{\text{dir}}(x; \theta_0, \phi_0, \kappa) = \kappa (R_o m(\theta_0, \phi_0))^{\top} (\widehat{x - t_o})$, where R_o maps from the anchor’s local to world frame, $(\widehat{x - t_o})$ is the unit direction from anchor to x , and κ is a system concentration parameter.

Composition and the role of the semantic kernel. The three log-likelihoods sum and are normalized over Ω_{free} . The semantic and metric kernels, however, are not independent: ℓ_{sem} is centred *on* the anchor while ℓ_{met} peaks *at* radius d_0 *from* it, so summing all three shrinks the composed mode inward. Along the modal ray ℓ_{dir} is constant in r , giving a closed form for the peak radius

$$r^* = d_0 \frac{\sigma_s^2}{\sigma_s^2 + \sigma_m^2} < d_0, \quad (1)$$

a systematic underestimate of the commanded offset. With σ_s scaled to the anchor extent ($\sigma_s = 0.5 \text{ max-dim}$) and $\sigma_m = 0.30 \text{ m}$, a 2m offset from a typical 0.8m fridge would peak at 1.28m, an error of 0.72m induced purely by the composition. We therefore treat the semantic kernel as an *anchor-selection* term rather than a goal-placement term. In

O-O mode it scores candidate instances and all three kernels apply. In O-W mode, where the goal is a free-space waypoint and the anchor is already resolved, the semantic term carries no information about *where along the offset the goal lies*, so we flatten it ($\sigma_s \rightarrow \infty$) and compose

$$\log P(x) = \ell_{\text{met}} + \ell_{\text{dir}}, \quad (2)$$

which by Eq. 1 recovers $r^* = d_0$ exactly. This is what makes the metric constraint hold to centimetre accuracy; naively multiplying all three experts would not.

VLM parameter inference. The Spatial Kernel Estimator prompts a VLM with the anchor label, current agent pose and yaw ψ , scene graph context, and an egocentric image. The VLM predicts the camera-relative azimuth θ_{cam} toward the goal region; the system converts this to a world-frame direction $\theta_0 = (\psi + \theta_{\text{cam}}) \bmod 2\pi$. For predicates classified as *intrinsic-** by the Orchestrator, θ_0 is further resolved using the anchor’s stored orientation rather than camera frame, enabling the commonsense subset’s frame-of-reference predicates. This design asks the VLM only to answer “in which camera direction is the goal?” (a task VLMs handle well) while the coordinate transform and metric enforcement are handled analytically.

4) Cascading Spatial Kernels. For compound instructions, one kernel triplet is instantiated per clause, each anchored to its resolved referent. All log-likelihoods are summed and renormalized, implementing a Product of Experts [23] whose peak satisfies the conjunction of all spatial constraints.

5) Verifier and Planner. The *Verifier* performs four checks on each proposed grounding: (i) orchestrator logic soundness, (ii) egocentric alignment (θ_{cam} consistent with what is visible in the image), (iii) semantic label flexibility (“sofa” accepted for “couch”), and (iv) absence of visual hallucination. A FAIL result appends structured feedback to the global failure history and re-triggers the Grounding Agent or Spatial Kernel Estimator as appropriate, forming a closed refinement loop. Waypoints that pass verification are forwarded to the *Planner* (RRT* [27]) for trajectory generation.

IV. DATASETS

Several benchmarks study spatial reasoning, including SpatialRGPT-Bench [15] and 3DSRBench [16], but their primary modality is single-image VQA, which precludes evaluation of (i) viewpoint consistency, (ii) partial observability, and (iii) the need for exploration before committing to an answer. Embodied QA benchmarks such as HM-EQA [28], used as an evaluation set by systems like GraphEQA [4], move closer by emphasizing interaction and persistent memory, but target semantic or factual questions rather than *metric-semantic-predicate* grounding (e.g., “2m left of the chair”) as an evaluation target.

To isolate 3D metric goal grounding, we introduce **MAPG-Bench** (Map-based Anchored Predicate Grounding Benchmark). MAPG-Bench contains 98 human-annotated queries across 41 HM3D [8] indoor scenes rendered in Habitat-Sim. It comprises two complementary query types:

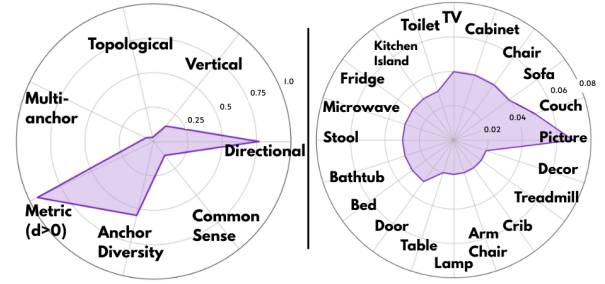


Fig. 5: **Distribution** of query categories (left) and anchor object labels (right) in MAPG-Bench. Example queries: (O-W, metric) “Where is 2m to the right of the fridge?”; (O-W, directional) “Where is directly behind the sofa?”; (O-O) “Which chair is nearest to the window?”; (commonsense) “Where is in front of the armchair?” (resolved via affordance direction).

Object-to-World (O-W) queries require localizing a continuous 3D allocentric waypoint relative to an anchor object (e.g., “Where is 2m to the right of the fridge?”), and **Object-to-Object (O-O)** queries require selecting the correct instance among multiple candidates satisfying a relational predicate (e.g., “Which chair is in front of the sofa?”).

Annotations were collected with an interactive tool that overlays semantic instance labels on the agent’s egocentric view. Annotators explored each scene, selected anchor instances, and recorded the agent’s pose. For O-W queries, the ground truth is an explicit 3D allocentric point placed on a navigable surface, an actionable waypoint rather than a text description. Predicates cover three families: horizontal directional (*in-front-of* 33, *right-of* 18, *behind* 13, *left-of* 11), vertical (*above* 10, *below* 4), and relational (*between* 6, *near* 1, *towards* 1, *from* 1). Metric distances span typical indoor scales (0.5–4.0m); a further 6 queries carry no metric offset ($d_0 = 0$) and serve as purely directional controls. A specialized *commonsense* subset probes VLM failures on intrinsic frame-of-reference predicates, where the “front” of an object is defined by its affordance direction rather than the camera’s line of sight, a systematically hard case documented in [7].

V. EXPERIMENTAL RESULTS

Task. We evaluate *goal grounding* under a common requirement: given a spatial query, egocentric observations, and a persistent 3D scene representation (scene graph), the system must output a grounded target that is directly usable by a planner. Depending on the query type, the output is either (i) an *allocentric waypoint* for object-to-world (O-W, “Where”) queries or (ii) an *object selection* for object-to-object (O-O) queries. Our evaluation isolates whether a method can (a) resolve the intended anchor under ambiguity, (b) interpret predicates in a consistent frame of reference, and (c) satisfy explicit metric constraints, while remaining robust under viewpoint changes and partial observability.

Metrics. On MAPG-Bench we report both geometric error

Method	d Err. O-O ↓ (m)	d Err. O-W ↓ (m)	Angle Err. ↓ (°)	Obj. Sel. ↑ (Acc.)	Common Sense ↑ (Acc.)	TSR ↑ (Rate)	Anchor Pick SR ↑ (Rate)	Avg. Traj. Len ↓ (m)
Open-Sourced Specialist								
SRGPT (SpatialRGPT-7B)	0.50	N/A	10.00	0.36	N/A	N/A	N/A	N/A
GraphEQA (Gemini 2.5 Pro)	N/A	5.82	30.78	0.34	0.14	0.78	0.48	1.32
Ours								
MAPG (Gemini 3 Pro)	0.58	0.08	15.13	0.30	0.14	0.46	0.46	14.60
MAPG (OpenAI GPT 5.2)	0.42	0.07	4.79	0.28	0.16	0.98	0.98	1.30
MAPG (Gemini 2.5 Pro)	0.08	0.45	6.19	0.42	0.08	0.90	0.94	2.50
MAPG (Claude Opus 4.6)	0.07	0.43	10.30	0.36	0.30	0.806	0.816	4.33

TABLE I: Comparison of MAPG against scene-graph EQA and single-image spatial-reasoning baselines on **MAPG-Bench**. Proprietary generalist models were benchmarked; they exhibit inconsistent performance and do not reliably produce actionable 3D goals across tasks, so we omitted them for space (available on our website). Metrics with ↑ are higher-is-better; ↓ are lower-is-better. **Obj. Sel.** measures whether the predicted target object satisfies the spatial query in O-O grounding, evaluated independently of navigation success. **Angle Err.** is the combined directional error between the predicted and ground-truth offset directions, computed as angle $\Delta = \arccos(\cos(\text{yaw err})\cos(\text{pitch err}))$ over the unit sphere. **TSR** measures end-to-end instruction completion, including both grounding and navigation. **Common Sense** measures accuracy on a specialized subset probing intrinsic frame-of-reference predicates (e.g., “in front of” resolved using object affordance direction rather than camera direction). **Anchor Pick SR** measures anchor instance disambiguation under partial observability. **Avg. Traj. Len** measures behavioral efficiency. †We did not run GraphEQA with Claude Opus 4.6 due to API cost constraints; this comparison is a direction for future work.

and task-level success. We measure distance error for O-O and O-W grounding (meters) and yaw/pitch angular error (degrees) to quantify directional consistency of the predicted offset. For O-O cases, we report object-selection accuracy using embedding-based soft matching between predicted and ground-truth object names to account for label-space mismatches between HM3D annotations and Hydra-derived semantic classes; MAPG is scored using the best match over its ranked Top-1/Top-2 predictions. For embodied efficiency, we report TSR, anchor-pick success rate (instance disambiguation under partial observability), and average trajectory length (meters).

Baselines. We compare MAPG against scene-graph EQA (GraphEQA) and single-image spatial specialists (SpatialRGPT). We modified GraphEQA’s planner interface to answer spatial “where” questions by producing a text waypoint from the scene graph, yielding a fair scene-graph-driven baseline for metric-semantic grounding. We evaluate multiple MAPG variants that share the same distributional grounding pipeline but differ only in the underlying VLM backend, isolating the effect of model choice from pipeline structure.

Completed end-to-end run. The completed evaluation used the Claude Opus 4.6 planner model, seed 42, the `bench_v1_98` split, and `llvmpipe` software rendering with `sim_gpu=-1`. All 98 episodes completed. MAPG succeeded on 79 episodes (80.6%), and the anchor was found on 80 (81.6%). Mean steps and mean trajectory length were both 4.33. The run made 830 model calls, including 142 retries, all of which were flagged; every call had a usage row, and total spend was \$15.780490 against a \$25 cap. During the run, an out-of-range Habitat semantic instance ID stopped question index 95; commit `c1b1ee5e` sanitized only out-of-range IDs, and the same run was resumed in place rather than

restarted after validating the seed, split, and configuration hashes.

Rendering fallback disclosure. The original run log contains 35 black-render fallback markers affecting split indices 14, 38, and 39. When every sampled image in an affected camera batch was filtered as black, the system used the final unfiltered raw image. Two of the three affected episodes succeeded. Excluding all three gives 77 successes in 95 episodes (81.1%), compared with 79 in 98 (80.6%) for the full result. This sensitivity check does not show that the fallback caused or prevented any outcome.

A) MAPG-Bench results: MAPG-Bench evaluates three grounding skills jointly: anchor instance disambiguation (Anchor Pick SR), predicate interpretation in a consistent allocentric frame (yaw/pitch error), and metric constraint satisfaction (O-W error). Table I reports quantitative performance. To illustrate the contrast, consider the query “Where is 2m to the right of the fridge?” GraphEQA selects the scene-graph node with the nearest bounding box satisfying the directional predicate, without enforcing the metric offset; the resulting waypoint is 5.82m from the ground truth. MAPG’s Orchestrator extracts {anchor: `fridge`, predicate: `right-of`, metric: `2.0m`}; the Grounding Agent resolves the fridge instance from the scene graph; the Spatial Kernel Estimator instantiates a vMF directional kernel ($\theta_0 = 90^\circ$ in the fridge’s local frame) and a radial Gaussian metric kernel ($d_0 = 2.0$ m, $\sigma_m = 0.30$ m); the composed distribution peaks 2.04m to the right of the fridge, yielding a waypoint 0.07m from the ground truth (see Fig. 4).

GraphEQA exhibits a large O-W distance error (5.82m) and high angular inconsistency (13.50° yaw, 27.90° pitch), reflecting its tendency to select the nearest region satisfying a predicate without enforcing the metric offset. Its Anchor

Predicate	n	Success
above	10	9 (90.0%)
behind	13	10 (76.9%)
below	4	1 (25.0%)
between	6	3 (50.0%)
from	1	1 (100%)
in front of	33	27 (81.8%)
left of	11	10 (90.9%)
near	1	1 (100%)
right of	18	16 (88.9%)
towards	1	1 (100%)

TABLE II: Success in the completed run, grouped by predicate.

Interpretation. Tables II and III are descriptive breakdowns of one split. *below* has 1 success in 4 examples (25.0%), and *between* has 3 in 6 (50.0%). These cells are hypotheses for follow-up, not stable conclusions. Several predicate and distance cells contain only one to three examples, so their percentages should not be generalized.

Pick SR (0.48) is also substantially lower than MAPG’s best (0.98), showing that committing early to a single scene-graph query under partial observability misses the correct instance roughly half the time. MAPG (OpenAI GPT-5.2) reduces O-W error from 5.82 m to 0.07 m (98.8% reduction, SEM 0.03 m), yaw error from 13.50° to 1.90° (85.9% reduction), and pitch error from 27.90° to 4.40° (84.2% reduction), while achieving 0.98 TSR and 0.98 anchor-pick SR with short trajectories (1.30 m). These gains follow directly from MAPG’s two design novelties: (i) explicit query decomposition (anchor / predicate / metric) and (ii) probabilistic composition of per-clause kernels into a planner-queryable goal distribution.

On O-O grounding, SpatialRGPT achieves 0.50 m error, consistent with its ability to reason about directional relations from a single image. However, when the target object is not visible in the best single view or requires integrating evidence across multiple viewpoints, SRGPT has no mechanism to resolve the discrepancy. MAPG (Claude Opus 4.6) reaches 0.07 m O-O error (86% reduction vs SRGPT), because the Grounding Agent accumulates multi-view evidence before committing to the anchor instance, and the Spatial Kernel Estimator composes a 3D belief anchored to the resolved object pose rather than a single-frame estimate.

This trade-off reflects the different demands of each task type: O-O grounding is primarily a disambiguation problem requiring accurate semantic matching and visual evidence accumulation across views, where Claude Opus 4.6’s strong CLIP-based referent resolution yields the lowest O-O error (0.07 m). O-W grounding additionally requires predicting precise metric offset parameters (d_0, σ_m) from a spatial predicate, where GPT-5.2’s superior instruction-following on structured spatial prompts produces better-calibrated metric kernels and the lowest O-W error (0.07 m). MAPG (Gemini 3 Pro) exhibits higher trajectory length and lower anchor-pick SR than MAPG (Gemini 2.5 Pro), consistent with Gemini 3 Pro being a preview checkpoint with weaker instruction-following on structured spatial prompts; we retain it for completeness. The overall pattern persists across all

Target distance	n	Success
0 m	6	3 (50.0%)
0.5 m	19	14 (73.7%)
1 m	34	28 (82.4%)
1.2 m	1	1 (100%)
1.5 m	12	10 (83.3%)
2 m	20	18 (90.0%)
2.5 m	3	3 (100%)
3 m	1	1 (100%)
4 m	2	1 (50.0%)

TABLE III: Success in the completed run, grouped by target distance.

Benchmarking on HM-EQA Question Answering		
Method	TSR $\uparrow$	L_τ (m) $\downarrow$
Explore-EQA	0.52	38.1
Explore-EQA-GPT4o	0.46	6.3
Explore-EQA-Llama4-Mav	0.44	10.4
Explore-EQA-Gemini-2.5Pro	0.54	12.3
GraphEQA-Gemini-2.5Pro	0.67	7.41
GraphEQA (Gemini 2.5 Pro)+	0.61	5.48
GraphEQA (GPT 5.2)+	0.63	N/A
MAPG (OpenAI GPT 5.2) (Ours)	0.60	N/A
MAPG (Claude 4.6 Opus) (Ours)	0.71	6.62

TABLE IV: EQA question-answering accuracy on HM-EQA [28]. + indicates our implementation of the particular baseline. MAPG achieves higher TSR while carrying out more exploration before committing to the target, consistent with its evidence-accumulation design. A GraphEQA + Claude Opus 4.6 comparison was omitted due to API cost constraints and is a direction for future work.

MAPG variants: low metric error and improved directional consistency versus the scene-graph baseline, validating that the *pipeline structure* (decomposition + composition) is the primary driver of spatial grounding performance.

B) HM-EQA Results (Generality without metric constraints): To assess whether MAPG’s metric-semantic decomposition degrades performance on tasks without metric constraints, we evaluate on HM-EQA [28], a semantic QA benchmark over HM3D scenes introduced by Explore-EQA where success is measured by multiple-choice answer accuracy over object properties and states complementary to MAPG-Bench’s continuous goal-grounding evaluation.

Table IV reports QA accuracy. Our method achieves improved or equivalent performance on HM-EQA, highlighting its generality and broad applicability even when metric constraints are absent. MAPG (Claude Opus 4.6) reaches 0.71 TSR, outperforming GraphEQA-Gemini-2.5Pro (0.67) and all Explore-EQA variants (0.44–0.54). This indicates that MAPG’s evidence-accumulation loop and anchor disambiguation, when decoupled from metric kernel prediction,

transfer effectively to semantic QA tasks. We use HM-EQA as a complementary diagnostic, while MAPG-Bench remains our primary evaluation for metric-semantic goal grounding.

C) Ablations: We ablate MAPG to quantify why explicit spatial reasoning and modular agents are necessary. Table V shows that removing the explicit spatial reasoner (replacing it with a single “fat” planner with curated chain-of-thought) drops object selection SR from 0.42 to 0.20 over the full $n = 98$ benchmark (Fisher exact $p < 10^{-3}$), while the full MAPG configuration improves over the GraphEQA base (0.34 to 0.42). These results support the claim that *explicit composition of metric and predicate constraints* is not recoverable from prompting alone, consistent with the running fridge example, where CoT-only reasoning misses the 2m offset entirely.

We further evaluate robustness under uncertainty induced by occlusion. In Table VI, the explicit spatial reasoner improves object selection SR from 0.30 to 0.50 under occlusion, suggesting that maintaining intermediate belief over anchors and deferring commitment can recover from partial observability when the anchor eventually enters the map. We report this as a qualitative indication rather than a statistical result: the occlusion subset contains only $n = 10$ queries over 5 scenes, so the difference amounts to two queries (Fisher exact $p = 0.65$) and is not significant. Enlarging this subset is ongoing work.

The commonsense-focused subset probes frame-of-reference predicates where the correct interpretation requires reasoning about intrinsic object orientation (e.g., the “front” of a chair is the side facing a seated occupant, not the camera’s front). On this subset (Table I, Common Sense column), MAPG (Claude Opus 4.6) achieves 0.30 accuracy, more than doubling the GraphEQA baseline (0.14). This gain reflects MAPG’s explicit predicate decomposition: rather than asking a single model to jointly resolve the referent and interpret the predicate, MAPG’s Grounding Agent first resolves the anchor instance and the Spatial Kernel Estimator separately reasons about the directional predicate in the anchor’s local frame. The remaining gap from perfect accuracy reflects the difficulty of intrinsic object-orientation inference from partial views, a target for future work in richer scene representations.

D) Real-World Demonstration: Using a Robotis AI Worker robot, we built an offline 3D scene graph from RGB-D observations of a physical lab environment and evaluated MAPG on three spatial queries, including “Where is 1 m to the right of the trash can near the bicycle?” MAPG resolved the correct anchor instance amid distractors, composed a spatial distribution peaking at the annotated target region (see Fig. 6), and produced a planner-ready waypoint within the expected tolerance. These results indicate that MAPG transfers to physical environments when a structured scene representation is available, though systematic real-world benchmarking is deferred to future work. A video of the demonstration is provided in the supplementary material.

Method	Object selection SR $\uparrow$
GraphEQA (base)	0.34
MAPG (CoT w/o spatial reasoner)	0.20
MAPG (explicit spatial reasoner)	0.42

TABLE V: Replacing the explicit spatial reasoner with CoT degrades object selection success rate, while the full MAPG variant achieves the best performance.

Method	Object selection SR $\uparrow$
GraphEQA (base)	0.30
MAPG (CoT- w/o spatial reasoner)	0.30
MAPG (explicit spatial reasoner)	0.50

TABLE VI: Occlusion highlights the limits of CoT-only reasoning, while the full MAPG variant performs best.

VI. OBSERVATIONS AND DISCUSSIONS

The completed Claude Opus 4.6 run succeeded on 79 of 98 episodes (80.6%) and found the anchor on 80 (81.6%). Tables II and III provide the corresponding breakdowns, with the small-cell caution stated alongside them. Tables I–VI show that MAPG’s gains stem from two design choices: **(i) query decomposition** (anchor / predicate / metric) and **(ii) probabilistic composition** of per-clause kernels into a planner-queryable goal density. On MAPG-Bench, MAPG (GPT-5.2) reduces O-W error from 5.82m to 0.07m and improves TSR from 0.78 to 0.98. Ablations in Tables V and VI show that these gains are primarily due to our compositional framework rather than “better prompting”: removing the spatial reasoner drops object-selection SR from 0.42 to 0.20, and under occlusion, explicit reasoning recovers SR from 0.30 to 0.50.

Multi-view evidence accumulation enables reliable anchor disambiguation. The Grounding Agent iterates over egocentric views, accumulates CLIP-scored evidence, and maintains a blacklist of Verifier-rejected candidates, deferring commitment until confidence exceeds threshold. This behavior directly explains MAPG’s high Anchor Pick SR (up to 0.98, vs. GraphEQA’s 0.48). GraphEQA selects the nearest scene-graph match at the first step; under partial observability this commits to the wrong instance roughly half the time. The occlusion ablation confirms this: when the anchor is initially occluded, MAPG’s iterative belief update recovers upon re-observation, whereas CoT-only reasoning has no mechanism to revise its initial commitment.

Log-space kernel composition yields planner-ready allocentric waypoints. MAPG’s final waypoint is the peak of a composed density over Ω_{free} : $\log P(x) = \ell_{\text{met}} + \ell_{\text{dir}}$ for O-W goals, with the semantic kernel acting on anchor selection rather than goal placement (Eqs. 1–2). Crucially, this is an *allocentric* 3D coordinate directly consumable by RRT* [27], with no interpretation step required. GraphEQA’s large O-W error (5.82m) and yaw error (13.50°) arise because its nearest-satisfying heuristic enforces direction but ignores the

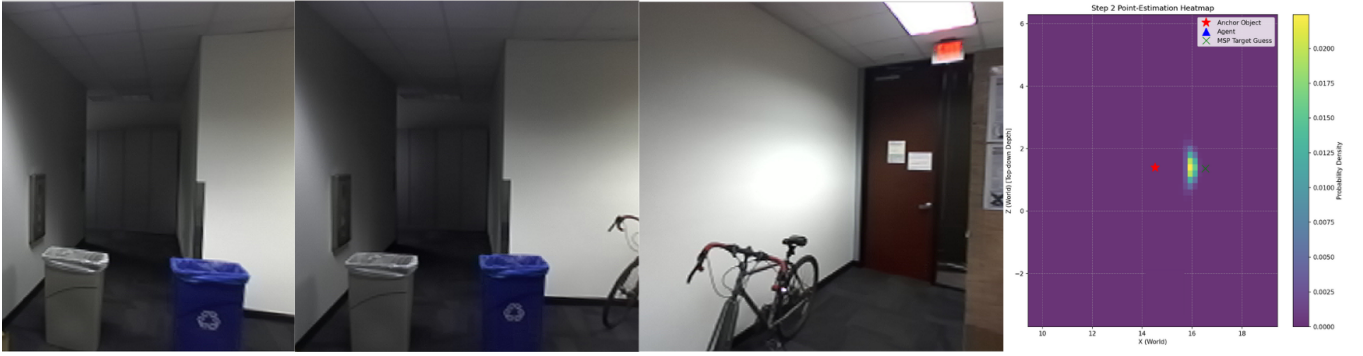


Fig. 6: **Real-world MAPG grounding example.** Using observations collected from a Robotis AI Worker robot, we construct an offline scene graph of a physical indoor environment. Three egocentric image frames are shown, with the query “Where is 1 meter to the right of the trash can close to the bicycle?” MAPG infers a spatial probability distribution over candidate goal locations (right). The anchor (red) and predicted goal (green) illustrate metrically consistent semantic grounding in a physical environment: the grounded area is 1 m to the right of the blue trash can, consistent with the bicycle’s position.

metric offset; MAPG’s metric kernel explicitly penalizes deviation from d_0 , collapsing the error to 0.07 m and 1.90° . MAPG’s equivalent results on HM-EQA (0.71 TSR vs. 0.67 for GraphEQA) confirm that the decomposition does not hurt when metric constraints are absent, while MAPG-Bench results confirm the full benefit emerges precisely when metric constraints and multi-predicate conjunctions stack together.

Failure modes and limitations. Most failures stem from limitations in the underlying scene representation. *Scene-graph incompleteness* (occluded objects that never enter the map) forces grounding relative to a partial world model. *Frame-of-reference ambiguity* is the dominant failure on the commonsense subset: for “in front of the sofa,” MAPG (Gemini 2.5 Pro) predicts the camera-centric front rather than the sofa’s affordance-defined facing direction, placing the waypoint behind the sofa, precisely the VLM bias documented in [7]. MAPG (Claude Opus 4.6) handles this better (0.30 vs. 0.14 for GraphEQA) because the Orchestrator’s `intrinsic_front` label triggers object-frame kernel computation, but performance still falls short of perfect, reflecting limits in inferring affordance direction from partial views. Finally, misalignment between the free-space grid and the scene-graph coordinate frame can smear the composed belief map, reducing waypoint precision for anchors near walls or furniture.

VII. CONCLUSION

In this work, we presented MAPG, a probabilistic grounding framework for translating metric-semantic language into planner-ready goals in continuous 3D Environments. Rather than treating grounding as a single hard decision, our approach decomposes a query into anchor, predicate and metric components, grounds them against an online 3D Scene Graph and egocentric observations, and composes their outputs into a continuous goal distribution. We also introduced MAPG-Bench, a benchmark designed specifically for metric-semantic goal grounding in embodied settings, covering 41 HM3D Scenes and 98 annotated queries. In the completed

98-query end-to-end evaluation, MAPG succeeded on 79 queries (80.6%) and found the anchor on 80 (81.6%). The predicate and distance breakdowns are uneven, and several cells are too small for stable conclusions. Our ablations further show that explicit decomposition and probabilistic composition contribute beyond prompting alone, and our real-world zero-shot demo illustrates the approach outside simulation. At the same time, failures arise from incomplete scene graphs, frame-of-reference ambiguity, and map alignment errors. Overall, this work argues that a distributional and compositional grounding approach provides a reliable interface between language understanding, spatial memory, and execution for open-world metric-semantic navigation.

REFERENCES

- [1] S. Tellex, N. Gopalan, H. Kress-Gazit, and C. Matuszek, “Robots that use language,” *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 3, pp. 25–55, May 2020, publisher Copyright: Copyright © 2020 by Annual Reviews. All rights reserve.
- [2] Y. Dai, R. Peng, S. Li, and J. Chai, “Think, act, and ask: Open-world interactive personalized robot navigation,” 2024.
- [3] S. Tellex, T. Kollar, S. Dickerson, M. R. Walter, A. Banerjee, S. Teller, and N. Roy, “Approaching the symbol grounding problem with probabilistic graphical models,” *AI Magazine*, vol. 32, no. 4, pp. 64–76, 2011.
- [4] S. Saxena, B. Buchanan, C. Paxton, B. Chen, N. Vaskevicius, L. Palmieri, J. Francis, and O. Kroemer, “GraphEQA: Using 3d semantic scene graphs for real-time embodied question answering,” *arXiv preprint arXiv:2412.14480*, 2024.
- [5] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Suen-derhauf, “Sayplan: Grounding large language models using 3d scene graphs for scalable robot task planning,” 2023.
- [6] J. Yang, S. Yang, A. W. Gupta, R. Han, L. Fei-Fei, and S. Xie, “Thinking in space: How multimodal large language models see, remember and recall spaces,” *arXiv preprint arXiv:2412.14171*, 2024.
- [7] Z. Zhang, F. Hu, J. Lee, F. Shi, P. Kordjamshidi, J. Chai, and Z. Ma, “Do vision-language models represent space and how? evaluating spatial frame of reference under ambiguities,” *arXiv preprint arXiv:2410.17385*, 2024.
- [8] S. K. Ramakrishnan, A. Gokaslan, E. Wijmans, O. Maksymets, A. Clegg, J. Turner, E. Undersander, W. Galuba, A. Westbury, A. X. Chang, M. Savva, Y. Zhao, and D. Batra, “Habitat-matterport 3d dataset (hm3d): 1000 large-scale 3d environments for embodied ai,” 2021.

- [9] A. Das, S. Datta, G. Gkioxari, S. Lee, D. Parikh, and D. Batra, "Embodied question answering," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [10] V. Cohen, J. X. Liu, R. Mooney, S. Tellex, and D. Watkins, "A survey of robotic language grounding: Tradeoffs between symbols and embeddings," 2024.
- [11] M. Ahn, A. Brohan, Y. Chebotar, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog, D. Ho, J. Ibarz, B. Ichter, A. Irpan *et al.*, "Do as i can, not as i say: Grounding language in robotic affordances," *arXiv preprint arXiv:2204.01691*, 2022.
- [12] C.-Y. Ma, Z. Wu, G. AlRegib, C. Xiong, and Z. Kira, "The regretful agent: Heuristic-aided navigation through progress estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [13] N. Hughes, Y. Chang, and L. Carlone, "Hydra: A real-time spatial perception system for 3d scene graph construction and optimization," in *Robotics: Science and Systems (RSS)*, 2022.
- [14] A. Rosinol, M. Abate, Y. Chang, and L. Carlone, "Kimera: an open-source library for real-time metric-semantic localization and mapping," 2020.
- [15] A.-C. Cheng, H. Yin, Y. Fu, Q. Guo, R. Yang, J. Kautz, X. Wang, and S. Liu, "Spatialrgpt: Grounded spatial reasoning in vision language models," *arXiv preprint arXiv:2406.01584*, 2024.
- [16] W. Ma, H. Chen, G. Zhang, Y.-C. Chou, J. Chen, C. M. de Melo, and A. Yuille, "3dsrbench: A comprehensive 3d spatial reasoning benchmark," 2025.
- [17] A. O'Neill *et al.*, "Open x-embodiment: Robotic learning datasets and RT-X models," *arXiv preprint arXiv:2310.08864*, 2023.
- [18] N. Hirose, C. Glossop, D. Shah, and S. Levine, "Omnivla: An omni-modal vision-language-action model for robot navigation," *arXiv preprint arXiv:2509.19480*, 2025.
- [19] C. Matuszek, "Grounded language learning: Where robotics and nlp meet," in *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence, IJCAI-18*. International Joint Conferences on Artificial Intelligence Organization, 7 2018, pp. 5687–5691.
- [20] L. Wang, J. Zhao, Y. Du, E. H. Adelson, and R. Tedrake, "Poco: Policy composition from and for heterogeneous robot learning," 2024.
- [21] O. Patil, P. Rath, K. Pangaonkar, E. Rosen, and N. Gopalan, "Factorizing diffusion policies for observation modality prioritization," *arXiv preprint arXiv:2509.16830*, 2025.
- [22] Y. Du and L. Kaelbling, "Compositional generative modeling: A single model is not all you need," 2024.
- [23] G. E. Hinton, "Training products of experts by minimizing contrastive divergence," *Neural Computation*, vol. 14, no. 8, pp. 1771–1800, 2002.
- [24] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," in *Forty-first international conference on machine learning*, 2024.
- [25] Y. Du, C. Durkan, R. Strudel, J. B. Tenenbaum, S. Dieleman, R. Fergus, J. Sohl-Dickstein, A. Doucet, and W. Grathwohl, "Reduce, reuse, recycle: Compositional generation with energy-based diffusion models and mcmc," 2024. [Online]. Available: <https://arxiv.org/abs/2302.11552>
- [26] T. Kollar, S. Tellex, D. Roy, and N. Roy, "Toward understanding natural language directions," in *Proceedings of the 5th ACM/IEEE International Conference on Human-Robot Interaction (HRI '10)*, 2010, pp. 259–266.
- [27] S. Karaman and E. Frazzoli, "Sampling-based algorithms for optimal motion planning," *The International Journal of Robotics Research*, vol. 30, no. 7, pp. 846–894, 2011.
- [28] A. Z. Ren, J. Clark, A. Dixit, M. Itkina, A. Majumdar, and D. Sadigh, "Explore until confident: Efficient exploration for embodied question answering," 2024. [Online]. Available: <https://arxiv.org/abs/2403.15941>